

Dynamically Heterogeneous Mesoscale Arrest in a Protein Polymer with Reversible β -Sheet-like Order

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Abstract

Dynamic arrest is commonly pictured as a progressive slowing of motion. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) of the sequence-defined elastin-like polypeptide (VPAVG)₃₀ at 30 °C resolves a different route. Over 84–131 min at that temperature, the fast-decay amplitude of a two-step correlation function falls from 0.38–0.70 to 0.00–0.13 across 76–167 nm, while the effective diffusion coefficient of that same fast mode changes by less than a factor of two. The mobile population

therefore empties far faster than it slows, and the largest length scales arrest first. Temperature-dependent SAXS/WAXS resolves a reversible transition to a hierarchical assembly carrying β -sheet-like peaks at $Q = 1.38$ and $0.72 \text{ \AA}^{-1}$ that vanish on cooling. Arrest by local conversion rather than bulk stiffening implies that a partially arrested protein network retains fully mobile regions, which bears on solute transport and on the maturation of biomolecular condensates.

Amyloid-like β -sheet ordering occupies an unusual position in protein science and polymer materials. Pathological amyloid fibrils formed by proteins and peptides that share no sequence homology are associated with Alzheimer’s disease,¹ Parkinson’s disease,² type II diabetes,³ prion diseases,⁴ and systemic light-chain amyloidosis.⁵ Many amyloid fibrils are kinetically persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-zipper interfaces.⁶ Yet amyloid-like order is not intrinsically pathological: functional amyloids store mammalian peptide hormones in pituitary secretory granules⁷ and build the extracellular matrix of bacterial biofilms,⁸ and related assemblies are increasingly used as mechanically robust biomaterials.⁹ What differs across these systems is not whether β -sheet-like order forms but how long it lasts: hormone granules disassemble to release functional monomer, whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a wide range of stabilities rather than committing a sequence to one permanent structure,¹⁰ and where a sequence falls in that range is a materials-design variable.

Reversible β -sheet-like contacts are attractive as candidate physical junctions because backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or weakly packed interfaces can remain dynamic or reversible.^{10–13} Elastin-like polypeptides (ELPs) are well suited to test this possibility because their sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly.^{14–18} Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.^{17–19} In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested

phase separation.^{20,21} Although these studies establish the arrested state from time-averaged scattering and rheology, how the assembly evolves toward arrest remains elusive.

Whether an assembled state is still evolving is a question for a dynamical measurement. In colloidal gels, microscopic relaxation continues to evolve after the time-averaged structure has nearly stopped changing, a nonequilibrium process termed physical aging.^{22,23} Related heterogeneous relaxation has been resolved during protein liquid–liquid phase separation,²⁴ protein gelation,²⁵ and cage relaxation in crowded protein solutions.²⁶ A protein-rich morphology can likewise keep reorganizing while its SAXS profile changes modestly, through junction exchange, whose kinetics set the network relaxation time in associative networks built from this same pentapeptide family,²⁷ and through the residual coarsening that follows arrested phase separation.^{28,29} A dynamical measurement can also separate two routes to arrest. If the assembly stiffens as a whole, relaxation times lengthen continuously. If it instead converts region by region, the material that is still mobile keeps its original relaxation rate while its share of the scattering falls. A bulk measurement averages the two together. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) probes the evolution directly through speckle decorrelation at the same low- Q length scales sampled by SAXS.

Here, temperature-dependent SAXS/WAXS and isothermal SA-XPCS provide complementary structural and dynamical views of concentrated aqueous (VPAVG)₃₀, an ELP built from 30 repeats of the pentapeptide valine-proline-alanine-valine-glycine. At 10 °C the scattering is dominated by molecular dimensions with no detectable mesoscale assembly; heating to 30 °C produces hierarchical low- Q scattering and high- Q peaks compatible with β -sheet-like packing, and cooling restores the low-temperature profile. In a separate experiment on a freshly loaded sample, isothermal SA-XPCS at 30 °C reveals a two-step decay whose fast fraction falls progressively with elapsed time. The mobile fraction therefore falls without the surviving dynamics slowing, while the local order that accompanies the assembly stays thermally reversible.

Figure 1 summarizes the SA-XPCS experiment. Its upper-left cartoon illustrates one can-

didate pathway, not a direct observation: protein-poor solvent regions appearing and growing within the slower protein-rich phase, coarsening it toward a network-like morphology. This dynamic-asymmetry picture follows Tanaka’s viscoelastic phase-separation framework.^{20,28} The (VPAVG)₃₀ sample was approximately 100 mg mL⁻¹ in the buffer given below. Each 2 s SA-XPCS acquisition at Advanced Photon Source (APS) beamline 8-ID-I (Experimental Section) used a previously unexposed sample position to limit radiation damage (SI Section 6); temporal and azimuthal averaging yielded $I(Q)$ and multi-tau correlation analysis yielded $g_2(Q, \Delta t)$.

Figure 2 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were measured at APS 12-ID-B by simultaneous SAXS/WAXS at 13.3 keV.³⁰⁻³² The “Measurement” aliquot had undergone more than 10 heating/cooling cycles; the “Reference” aliquot had stayed at 4 °C. At 10 °C the two overlap, and a Guinier fit to the Reference gives an apparent $R_g = 22.65 \pm 0.50$ Å (Figure S3), a dispersed state with no mesoscale aggregates down to $Q = 0.0081$ Å⁻¹, which corresponds to structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(Q) \propto Q^{-3.2}$ response at low Q (slopes -3.17 and -3.15 over $0.012 < Q < 0.040$ Å⁻¹). Both also develop the same pair of high- Q features, at $Q = 1.38$ and 0.72 Å⁻¹ in the Reference and 1.36 and 0.74 Å⁻¹ in the Measurement. The cycled aliquot alone makes the point: it still develops both peaks at 30 °C and shows neither at 10 °C. A more direct test at 8-ID-I took a single aliquot through seven consecutive 6–34 °C cycles, heated at 1 °C min⁻¹ (Figure S4). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation with no systematic drift, and at 33.8 °C the correlation function stays close to $1 + \beta$ in every cycle, the signature of an arrested state. The sample therefore assembled and arrested on every heating and returned to the same low-temperature state on every cooling (SI Section 7). Video S1 independently shows recovery of optical transparency after cooling.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle scattering signifies density heterogeneity extending across the probed interval rather than a



Figure 1: SA-XPCS measurement of thermally induced $(\text{VPAVG})_{30}$ assembly at APS 8-ID-I. Each 100,000-frame set of sparse $20\ \mu\text{s}$ frames feeds two analyses. Averaging them in time gives the SAXS intensity $I(Q)$: how much structure the sample has at each length scale $2\pi/Q$, which grows at low Q on heating. Correlating them in time gives $g_2(Q, \Delta t)$: how fast the structure at that same length scale rearranges. A decay in g_2 means the density fluctuations at $2\pi/Q$ are still moving; a flat g_2 means they are frozen over the accessible delay times Δt . Both are sketched beside the detector (red: assembled; blue: dispersed; illustrative, not data). Center: the aluminum cell, liquid chamber 3 mm across. The photos, outlined in blue and red, show the ELP sample below and above the transition. Lower left: one $(\text{VPAVG})_{30}$ repeat, residues colored as keyed. Right: the frame-averaged detector image, color bar in photons per pixel per frame, white center the beamstop mask; the graininess is sparse counting statistics, not resolved speckle. Upper left, stages i–iv: a conceptual illustration of the viscoelastic phase-separation pathway,^{20,28} in which protein-poor solvent pools (cyan) nucleate and grow within the still-continuous protein-rich phase (deepening red: more protein), coarsening it toward a network-like morphology.

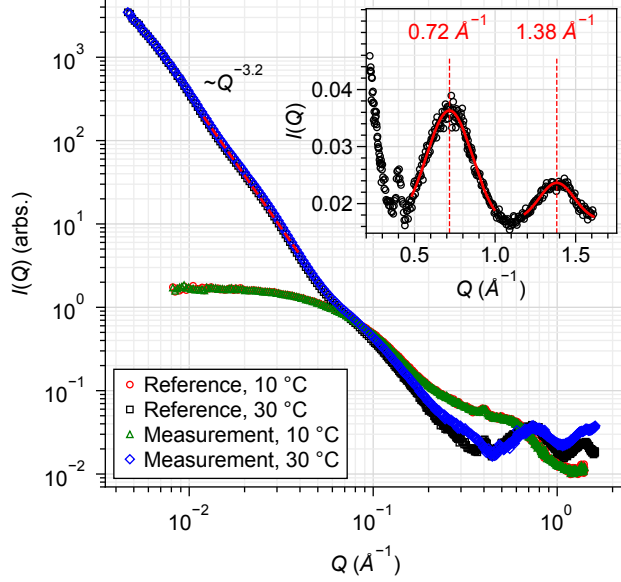


Figure 2: Temperature-dependent structure and reversibility of $(\text{VPAVG})_{30}$. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50 \text{ \AA}$ (Figure S3). Heating produces an approximate $Q^{-3.2}$ low- Q response and peaks at $Q = 1.38$ and $0.72 \text{ \AA}^{-1}$, corresponding to real-space spacings of approximately 4.5 and 8.8 $\text{\AA}$, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < Q < 0.040 \text{ \AA}^{-1}$. Inset: the high- Q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70 \text{ \AA}^{-1}$, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

single resolved structural scale.³³ Related gel-forming ELPs exhibit disordered bicontinuous or fractal protein-rich networks produced by arrested phase separation.^{20,21} The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross- β -like assemblies, whose reflections are reported at approximately 4.8 Å along the fibril direction and between 8 and 11 Å across it.^{6,34} Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 Å⁻¹ (correlation lengths $2\pi/\Delta Q$ of only about 25 and 19 Å), so the ordering extends over a few repeat spacings rather than forming extended cross- β crystallites. Their concurrent appearance with the low- Q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity. The disappearance of the peaks on cooling identifies thermally labile β -sheet-like associations that are plausible, rather than uniquely established, physical junctions.

To resolve mesoscale arrest isothermally, a freshly loaded sample was held at 6 °C for more than 30 min, warmed to 20 °C at 5 °C min⁻¹, then to 30 °C at 1 °C min⁻¹ to limit overshoot. The waiting time t_w starts when the calibrated resistance-temperature detector (RTD) at the cell read 30 °C, and simultaneous SAXS and SA-XPCS then ran for 7863 s (Figure 3). Each t_w labels a group of averaged consecutive acquisitions at previously unexposed positions (SI Section 2.2). The SAXS intensity starts close to the 6 °C reference and rises by more than two orders of magnitude at low Q as t_w advances, staying broad and monotonic (Figure 3a). The profiles stay broad and featureless as they rise, with no peak emerging, so the assembly gains density contrast without settling on a single characteristic domain size. Before $t_w = 5040$ s the low- Q coherent intensity is too weak to resolve a two-step decay above the noise, and above $Q = 0.00827$ Å⁻¹ the scattered intensity has fallen by more than an order of magnitude (Figure 3a) so g_2 is dominated by counting noise. Reliable correlation functions are therefore restricted to $t_w \geq 5040$ s and the five lowest Q bins, where the measurements were averaged in time to improve the statistics.

The intensity autocorrelation function g_2 for a single SA-XPCS acquisition is defined as:

$$g_2(Q, \Delta t) = \frac{\langle I(Q, t)I(Q, t + \Delta t) \rangle_t}{\langle I(Q, t) \rangle_t^2}, \quad (1)$$

where $I(Q, t)$ is the coherent scattering intensity and Δt is the delay time. Through the Siegert relation, g_2 reports the relaxation of density fluctuations at length scale $2\pi/Q$. At $Q = 0.00376 \text{ \AA}^{-1}$ (approximately 167 nm), the correlation functions show a two-step decay: an initial millisecond-scale relaxation followed by an apparent plateau that persists throughout the 2 s acquisition (Figure 3b). The fast decay was fully resolved using the Rigaku XSPA-500k detector with a 50 kHz continuous frame rate.³⁵

As t_w elapses over repeated acquisitions, the amplitude of the first decay decreases and the plateau rises. Such two-step relaxation and growth of a slow component are characteristic of heterogeneous kinetic arrest in protein and colloidal gels.^{22,25,36} The two-step decays in g_2 were parameterized by:

$$\begin{aligned} g_2(Q, \Delta t, t_w) &= 1 + \beta |g_1(Q, \Delta t, t_w)|^2, \\ g_1(Q, \Delta t, t_w) &= f(Q, t_w) \exp[-(D_0(t_w) Q^2 \Delta t)^{p_{\text{fast}}}] \\ &\quad + [1 - f(Q, t_w)] \exp\left[-\left(\frac{\Delta t}{\tau_{\text{slow}}(Q, t_w)}\right)^{p_{\text{slow}}(t_w)}\right], \end{aligned} \quad (2)$$

The first line is the Siegert relation, in which g_1 is the normalized intermediate scattering function and β is the instrumental speckle contrast, measured on a static reference and held fixed (Figure S7). The second line is the two-mode model, where f is the fitted fast fraction and τ_{slow} an effective slow relaxation time. The effective fast relaxation time is replaced by $1/(D_0 Q^2)$ in the fast term, where D_0 is a fitted diffusion coefficient. The stretching exponents p_{fast} and p_{slow} describe the breadth of the relaxation-time distributions. Note that a stretched correlation function can still be diffusive as long as p_{fast} does not depend on Q and the characteristic time scales with Q^{-2} . The effective diffusion coefficient is calculated using the

mean XPCS relaxation time defined in:³⁷

$$\langle \tau \rangle = \frac{\Gamma(1/p)}{p} \tau, \quad D_{\text{eff}}(t_w) = \frac{1}{\langle \tau_{\text{fast}} \rangle Q^2} = \frac{p_{\text{fast}}}{\Gamma(1/p_{\text{fast}})} D_0(t_w), \quad (3)$$

Fitting p_{fast} once for the whole series rather than at each elapsed time separately gives 0.595 ± 0.022 . τ_{slow} exceeds the acquisition window, so it is an extrapolated descriptor; the primary quantities are f and D_{eff} .

Figure 3c shows that the fitted fast fraction decreases monotonically with elapsed time at every measured Q . At fixed $t_w = 5040$ s, f increases with Q , from 0.38 at $Q = 0.00376 \text{ \AA}^{-1}$ (167 nm) to 0.70 at $Q = 0.00827 \text{ \AA}^{-1}$ (76 nm); by $t_w = 7863$ s the same bins give 0.00 and 0.12 (Figure 3c), so the complementary arrested fraction $1 - f$ has gone from 0.62 and 0.30 at those two Q values to 1.00 and 0.88 over approximately 47 min. An increasing fraction of the scattering ensemble is therefore arrested within the accessible delay range, at every length scale probed. At every analyzed time f also increases monotonically with Q , end to end across the five bins by 9.0 standard deviations or more (Figure 3c and SI Section 5.3), so arrest develops first and most strongly at the largest length scale probed, 167 nm, and progressively less so toward 76 nm. That ordering holds within each elapsed-time group, so it reflects length scale rather than the trend with t_w , and it ties the growing low- Q SAXS structure to the arrest fraction measured on the very same frames. Note that XPCS does not locate the arrested regions, and the scattering contributions are weighted by structure and contrast, therefore $1 - f$ is not a geometric volume fraction.

Figure 3c also implies what survives in the mobile population is ordinary diffusive material. D_{eff} rises and then falls across the series, spanning 1.8 to $5.3 \times 10^7 \text{ \AA}^2 \text{ s}^{-1}$, ending a factor of 1.8 below where it started, a difference of 2.9 standard deviations (Figure 3c). Progressive stiffening would look different: as regions tightened, their density fluctuations would relax more slowly and would stop relaxing diffusively. Instead the surviving fraction keeps a diffusive Q^{-2} relaxation at a coefficient that moves by less than a factor of two while the fraction itself falls to zero. What changes systematically is therefore how much material

relaxes quickly, not how quickly it relaxes. Across the same interval f falls by a factor of five at the smallest length scale and to zero at the largest. Both the amplitude and the rate change, but the amplitude changes further and does so monotonically, so it is the amplitude that carries the arrest.

The same pairing of a plateauing diffusive fast time scale with a declining fast amplitude was found in a thermoreversible nanoparticle gel.²² Both stretching exponents stay below unity at every elapsed time, the signature of a broad distribution of relaxation times rather than a single process; the product of p_{fast} with the Q exponent of -2 is 1.2 rather than the value of 2 that subdiffusive motion of identical scatterers would give.³⁷ The dynamic heterogeneity in similar colloidal gel systems have been confirmed using both confocal microscopy and MD simulation.^{29,38}

One thing to note is that the fast amplitude and the fast relaxation time are not equally well determined at the last elapsed time: f is set by the height of the plateau and remains measured, whereas $\langle\tau_{\text{fast}}\rangle$ needs a resolved decay and is lost once f falls to a few percent. At the largest t_w the fitted amplitude in the lowest- Q bin refines to zero. Equation 2 reduces to its slow term there, so that correlation function is described by the slow component alone and places no constraint on D_0 . D_{eff} at that waiting time is therefore set by the four higher- Q bins. τ_{slow} is not used for mechanistic scaling because it lies beyond the acquisition window (Figure S8). Correlation functions at the remaining Q values are shown in Figure S9. Fresh material for every acquisition keeps cumulative dose from tracking t_w , so the trend in f cannot be a dose artifact, and a control at 23–41 times this flux shows no change in g_2 with dose rate, which rules out radiation-induced artifacts (Figure S10).

The growth of low- Q SAXS intensity and the decline of the fast fraction are two complementary aspects of network formation. This coupling is consistent with network-forming arrested phase separation in a dynamically asymmetric protein–solvent mixture, where concentration fluctuations build a connected dense phase and viscoelastic constraints suppress relaxation.^{24,25,28,39} Related ELPs form bicontinuous networks with stress-relaxation times

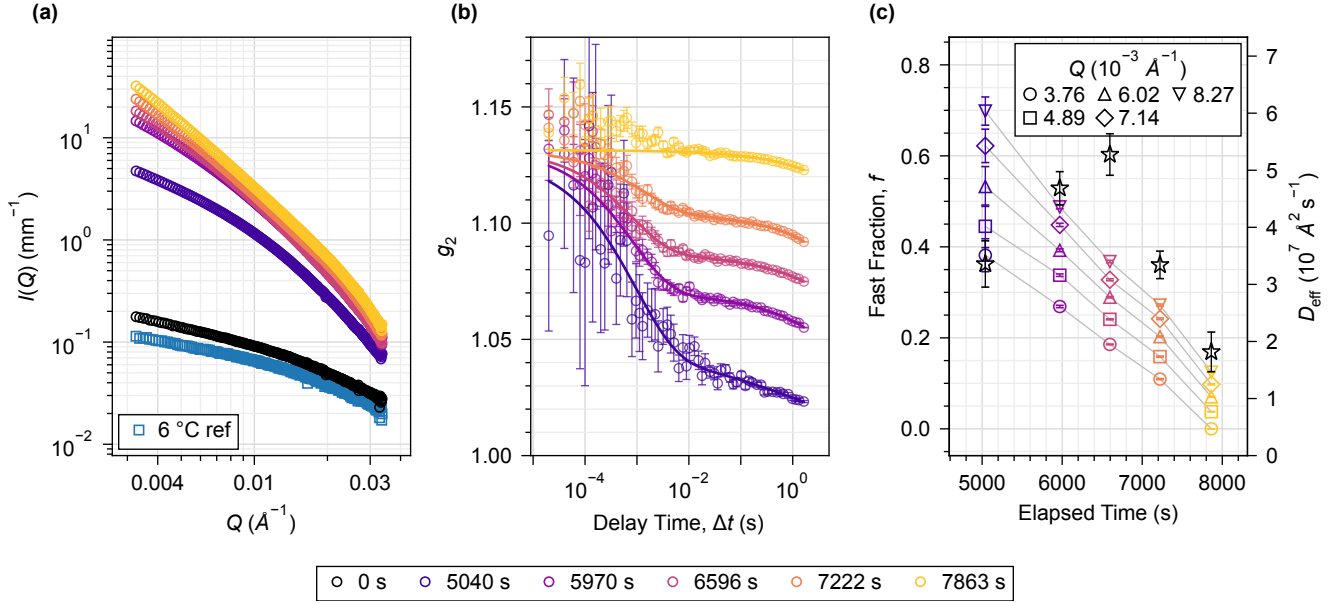


Figure 3: Isothermal structure evolution and dynamic arrest of (VPAVG)₃₀ at 30 °C. (a) Buffer-subtracted SAXS on an absolute scale (absolute scattering cross section per unit volume, calibrated as in Figure S6), showing continued growth of low- Q density contrast. Panel (a) alone also carries the 6 °C reference (blue squares, keyed separately) and the 0 s profile; (b) and (c) cover the five later times only. Full series in Figure S5. (b) Representative $g_2(Q, \Delta t)$ at $Q = 0.00376$ Å⁻¹; solid lines are global two-mode fits, and the error bars are the measured pointwise g_2 uncertainties. With elapsed time, the amplitude of the first decay decreases and the persistent plateau rises. Correlation functions and fits at the four other Q values are given in Figure S9. (c) Left axis, open coloured markers: the fitted fraction of fast dynamics, f , decreases at all five Q values; grey lines connect points at the same Q and are guides to the eye. Right axis, open black stars: the effective diffusion coefficient of the fast mode, $D_{\text{eff}} = [p_{\text{fast}}/\Gamma(1/p_{\text{fast}})] D_0$ from Equation 3, one fitted value per elapsed time. The amplitude falls monotonically to zero; the rate rises and then falls through a factor of three. All error bars are one-standard-deviation fit uncertainties. Slow-mode parameters are given in Figure S8. Colors denote elapsed time throughout, keyed along the bottom; each elapsed time labels a group of averaged acquisitions as defined in the text.

beyond 10^3 s.^{20,21} The present measurements add a local structural element from a separate experiment: at high temperature (VPAVG)₃₀ carries thermally reversible β -sheet-like contacts. The separate experiments cannot place the ordered contacts at the junctions, but the two are compatible in mechanism. Arrest that empties the mobile population rather than slowing it is what cooperative junctions would produce: a region leaves that population once enough contacts form together, rather than relaxing ever more slowly as each one tightens. β -sheet assembly is cooperative in exactly that sense: it must clear a nucleation barrier before growth begins,⁴⁰ it accelerates once started,⁴¹ and it extends by templated addition onto a self-complementary interface.⁶ Structure and dynamics therefore point to the same kind of local process.

The route to arrest matters for what a partially arrested network can do. A medium that stiffens throughout slows down every solute within it. One that converts region by region keeps pockets that are not merely less viscous but still diffusive, so transport should follow the connectivity of those pockets rather than a bulk viscosity, and release should hold and then fall at a threshold rather than tail off. Because the arrested fraction evolves in time and can be measured, a network interrupted at a chosen elapsed time carries a defined fraction, a handle that composition alone does not give. The same question is open for biomolecular condensates, whose maturation from liquid to solid governs their function and their role in aggregation.^{42,43} If maturation proceeds the same way, a partially matured condensate keeps interior regions that still exchange, and the liquid-to-solid transition is a connectivity threshold rather than a gradual drift. Condensates are too small and too dilute for the measurement reported here. A concentrated, sequence-defined mimic is not, and in this one arrest is local.

Four limitations bound the interpretation, and each points to the measurement that would lift it. First, the β -sheet-like assignment rests on WAXS peak positions rather than spectroscopy; attenuated-total-reflection infrared through a thermal cycle is the natural test, circular dichroism being poorly suited to a sample this turbid. Second, the high- Q

ordering and the low- Q dynamics come from separate experiments, so no time-resolved causal relation between them is established; simultaneous WAXS and SA-XPCS on one loading would settle it. Third, the 2×10^{-5} –1.6 s delay range quantifies the fraction arrested on that scale rather than the infinite-time nonergodicity factor, for which linear shear rheology on a scaled-up preparation is the complement. Fourth, the isothermal series comes from one loaded sample, with fit uncertainties rather than between-sample variability, so the trend in f is a single-preparation observation.

Subject to those constraints, the two measurements give one picture. SA-XPCS resolves how the assembly arrests: the mobile population stays diffusive throughout, its effective diffusion coefficient changing by less than a factor of two while its amplitude falls by a factor of six at the smallest length scale and to zero at the largest, so arrest advances by converting material rather than by slowing it, and it reaches the largest length scales first. SAXS/WAXS resolves what the assembly is made of: a hierarchical network carrying β -sheet-like local order that vanishes on cooling. Put together, a protein network can arrest region by region while the contacts that build it stay thermally erasable, which is what makes amyloid-like packing a candidate switchable junction rather than a commitment to permanent aggregation.

Experimental Section. The Sanger-verified construct encoded an N-terminal His-tagged (VPAVG)₃₀ protein (complete sequence in the Supporting Information). Recombinant (VPAVG)₃₀ was expressed in *E. coli*, purified by immobilized metal-affinity chromatography, and buffer exchanged into 20 mmol L⁻¹ HEPES, 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT at pH 8. Expression, solubility, and purity were monitored by SDS-PAGE (Figure S1b). Protein concentration was determined from A_{280} using the sequence-specific extinction coefficient ($\epsilon_{280} = 1490 \text{ M}^{-1} \text{ cm}^{-1}$, 15,461 Da construct); the final concentration was approximately 100 mg mL⁻¹.

Temperature-dependent SAXS/WAXS measurements were performed at APS 12-ID-B with a 13.3 keV beam of approximately $100 \times 150 \mu\text{m}^2$ and an incident flux of approximately 5×10^{11} photons s⁻¹. At each set point the capillary was held at temperature for at least

10 min before exposure. SAXS and WAXS were acquired simultaneously in 0.1 s exposures with an Eiger2 9M at approximately 2 m and a Pilatus 300K at 0.42 m. SA-XPCS was performed at APS 8-ID-I with a 10.94 keV, $10 \times 10 \mu\text{m}^2$ partially coherent beam and a Rigaku XSPA-500k detector 7.8 m downstream. Each acquisition contained 100,000 frames at 50 kHz (20 μs per frame, 2 s total). Further acquisition, fitting and radiation-control details are in the Supporting Information.

Supporting Information

Recombinant synthesis and purification; gene-construction sequences and SDS-PAGE analysis; sample environments; SAXS/WAXS calibration; SA-XPCS analysis and global fitting; speckle-contrast calibration; radiation controls; Guinier analysis; complete isothermal SAXS evolution; length-scale-dependent XPCS parameters; correlation functions at all measured Q values; thermal-cycling repeatability; a still frame and link for the video of the reversible thermal transition; code and reduced-data availability; and AI-assisted development (PDF).

Data and Code Availability

Reduced data and analysis/figure-generation code are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>. The repository includes the 8-ID-I and 12-ID-B datasets used for the figures, radiation-control data, and analysis scripts, and every data-derived figure can be regenerated from it alone. Raw beamline data are available from the corresponding authors upon reasonable request.

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AI-assisted manuscript and code preparation. The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and with development of the data-analysis and figure-generation code. The Claude models were accessed through Argo, Argonne National Laboratory’s internal generative AI interface (see Supporting Information). That code and the reduced data it operates on are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>, so the reported results can be regenerated and checked independently. The authors take full responsibility for all manuscript content and reported results.

Conflict of Interest

The authors declare no competing financial interest.

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